DAURES GREEN HYDROGEN VILLAGE

University of Stuttgart
IER Institute of Energy Economics and Rational Energy Use

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Life-Cycle Assessment and Hydrogen Certification

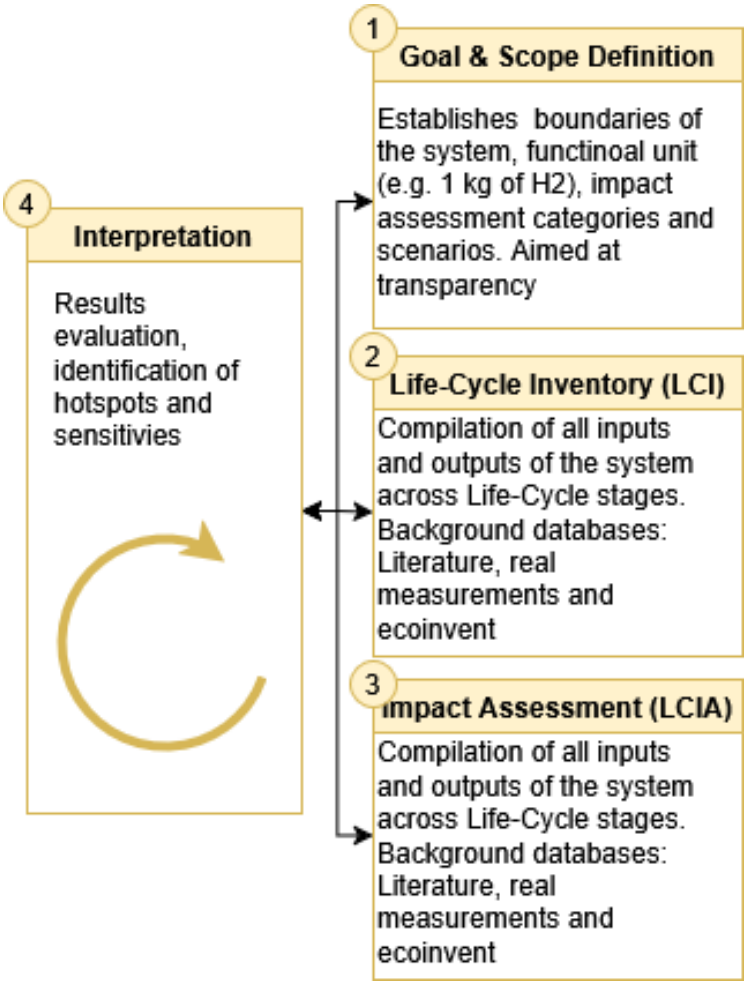
OVERVIEW

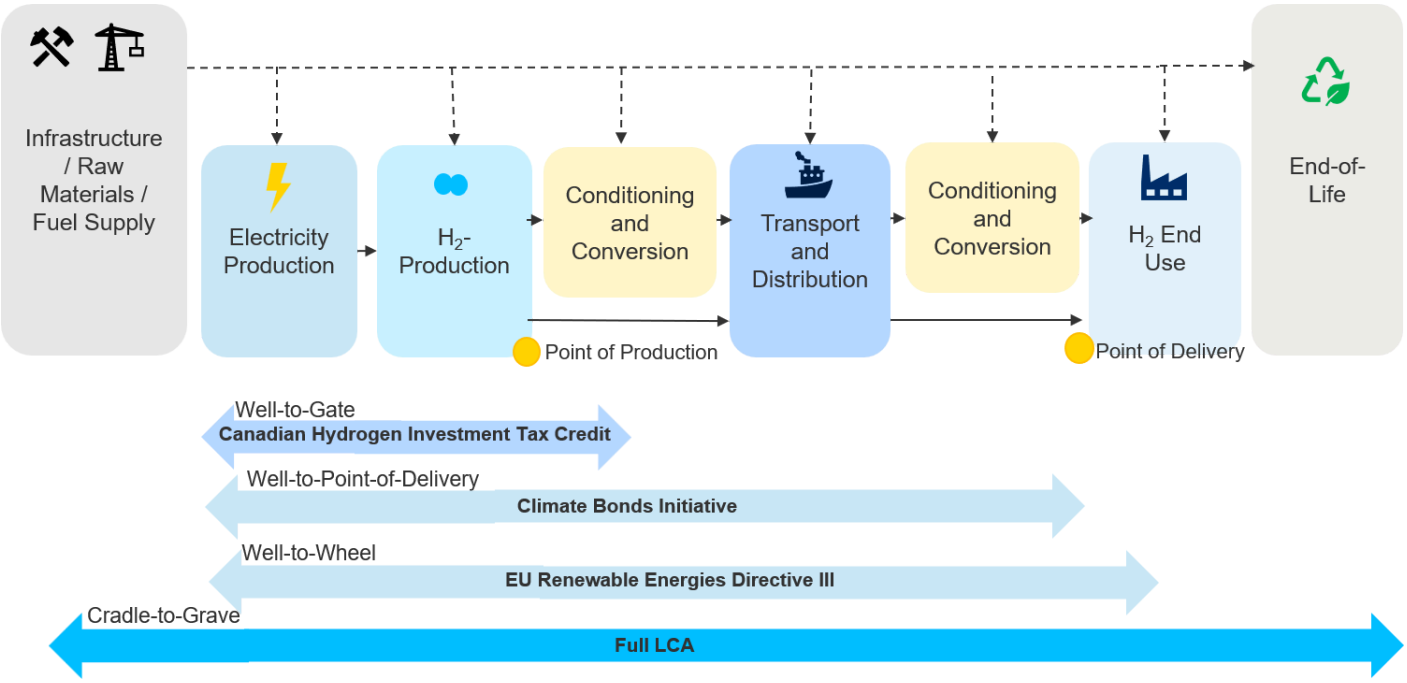
Life-Cycle Assessment (LCA) is an internationally standardized methodology to assess the environmental impacts across the life-cycle of a product, process or service. With regards to efuels and hydrogen certification, it has become the key tool to quantify the greenhousegas (GHG) emissions and ensure compliance with regulatory thresholds.

FUNDAMENTALS

LCA is standardized in the ISO 14040/44 series [1], the approach follows the scheme shown to the right of the page. A full LCA considers all stages from cradle-to-grave of the system under analysis. In regulatory schemes LCA, these are typically cut off at the production- or delivery-gate to focus on these Life-Cycle Stages specifically. Below, the system boundaries of hydrogen production systems using electrolysis are shown schematically for the Canadian Hydrogen Investment Tax Credit, the Renewable Energies Directive III by the EU, the voluntary Climate Bonds Initiative Standard as well as full LCA [2-4].

LCA FRAMEWORK (ISO 14040/44)

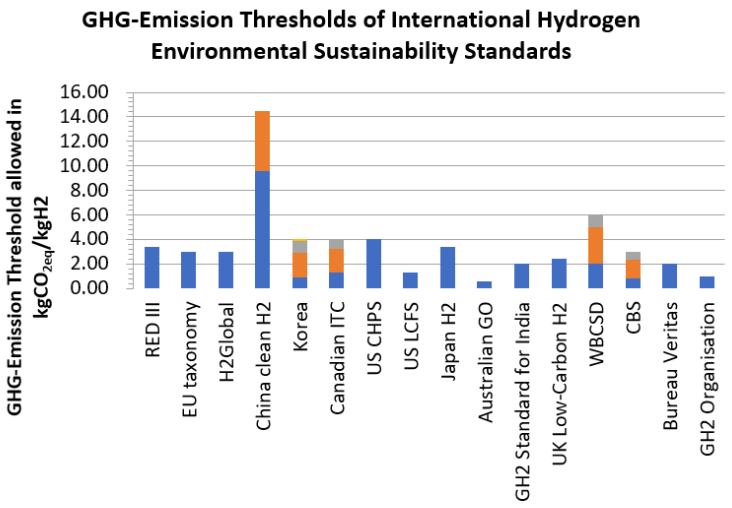




H2 STANDARDS ANALYSIS

An overview over four different standards regulating the sustainability of hydrogen production is given in the figure below. The current regulatory landscape for hydrogen shows significant differences in terms of scope, eligible technologies, sustainability criteria and system boundaries along the Life-Cycle as well as GHG emission thresholds allowed.

Notably, the GHG emission threshold varies between 0.75 and 14.90 kgCO2/kgH2, with many standards adopting the criteria and methodology of the Renewable Energies Directive III legislated by the EU. Beneath GHG-emissions, the sourcing of the renewable electricity for electrolysis (the only viable technology under that scheme) must correlate to its consumption temporaly as well as geographically. This counts alos towards imported hydrogen from outside the EU.



At IER, we conduct thorough Life-Cycle Assessment according to ISO standards, while also considering future changes in the energy system, which will have implications for the environmental impacts.

Standard	Type	Carbon Threshold	Unit	Specification	Technology open?	Additional Criteria?	Derivatives?
Renewable H2 EU RED III	regulatory	3.38	kgCO2eq/kgH2	N/A	n	Geographical & Temporal Correlation + Additionality	e-fuels
Canada Clean Hydrogen Investment Tax Credit	regulatory	<0.75 - < 4.0	kgCO2eq/kgH2	99.9 vol% purity, 1-30 bar	y	Labour Conditions	Ammonia, Methanol, e-fuels
H2Global	Incentive	3	kgCO2eq/kgH2	99,0 vol% purity, 30 bar	n	95 % RES + other environmental & social criteria	Ammonia, Methanol, Synthetic Natural Gas
Climate Bonds Initiative	voluntary	3.00 and decreasing	kgCO2eq/kgH2	99,9 vol% purity, 30 bar	y	Pollution, Land & Water Use	Ammonia, Methanol, Liquid Organic H2 Carriers

REFERENCES

[1] ISO 14040/44, The Life-Cycle Assessment Methodology
[2] Renewable Energies Directive III, European Commission 2023
[3] Canada Clean Hydrogen Investment Tax Credit, Government of Canada
[4] Climate Bonds Initiative 2022

References used in table and graph

H2Global Foundation 2022, The Green Hydrogen Organisation GH2 Standard 2023, Climate Bonds Initiative 2021, Liu et al 2022, Bureau Veritas 2024, US Clean H2 Production Standard, UK Low Carbon Standard, Australian GO scheme

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https://github.com/IER-Hy4Daures/Fact-Sheets

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